ASSIGNMENT -2

Multiclass Image Classification Using Convolutional Neural Networks and Transfer Learning

Course

Deep Learning Architectures and Techniques

Assignment

Practical Lab Assignment 1

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1. Introduction

Image classification is one of the most important applications of computer vision and deep learning. It involves automatically identifying and categorizing objects within an image into predefined classes. Traditional machine learning techniques required manual feature extraction, which often limited performance. However, deep learning methods, particularly Convolutional Neural Networks (CNNs), have significantly improved image classification accuracy by automatically learning spatial features from raw images.

Convolutional Neural Networks are specialized neural networks designed to process grid-like data such as images. CNNs utilize convolution operations, pooling layers, and nonlinear activation functions to extract hierarchical patterns from images. These networks can identify low-level features such as edges and textures as well as higher-level features such as shapes and objects.

In this experiment, a CNN model is implemented to perform multiclass image classification using the Fashion-MNIST dataset. Two approaches are explored:

1. A custom Convolutional Neural Network architecture designed from scratch.
2. A pretrained CNN model using transfer learning.

The objective of this assignment is to understand how CNNs work for image classification tasks and how transfer learning can improve performance by leveraging pretrained models.

2. Dataset Description

The dataset used in this experiment is the **Fashion-MNIST dataset**, which is a widely used benchmark dataset for image classification. It was developed as a replacement for the MNIST handwritten digit dataset and consists of grayscale images of clothing items.

Dataset Characteristics

Training Images: 60,000
Testing Images: 10,000
Image Resolution: 28 × 28 pixels
Number of Classes: 10

Each image in the dataset belongs to one of the following categories:

1. T-shirt/top
2. Trouser
3. Pullover
4. Dress
5. Coat
6. Sandal
7. Shirt
8. Sneaker
9. Bag
10. Ankle Boot

Each image is represented as a 28×28 grayscale image. The task is to correctly classify each image into one of the ten categories. Since there are multiple classes involved, this problem is classified as a **multiclass classification problem**.

3. Image Preprocessing

Before training the CNN model, several preprocessing steps were applied to prepare the dataset.

3.1 Normalization

The pixel values in the dataset range from 0 to 255. To improve training stability and speed, these values were normalized to the range between 0 and 1 by dividing each pixel value by 255.

Normalization helps neural networks converge faster during training and improves numerical stability.

3.2 Reshaping Images

Convolutional Neural Networks require input images in the format:

Height × Width × Channels

Since the Fashion-MNIST images are grayscale, they contain only one channel. Therefore, the dataset was reshaped to include the channel dimension.

3.3 Train-Test Split

The dataset already contains predefined training and testing sets:
Training Set: 60,000 images
Testing Set: 10,000 images

The training dataset was used to train the model, while the testing dataset was used to evaluate the model's performance.

4. Data Augmentation

Data augmentation is a technique used to artificially increase the size and diversity of the training dataset. It helps improve model generalization and reduces overfitting.

The following augmentation techniques were applied using the ImageDataGenerator class:

• Image rotation
• Width shifting
• Height shifting
• Zoom transformations

These transformations create modified versions of the training images during the training process. As a result, the model learns to recognize objects even when they appear in slightly different orientations or positions.

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5. Custom CNN Architecture

A Convolutional Neural Network was implemented from scratch to perform image classification.

The architecture of the CNN consists of the following layers:

Convolutional Layer 1

Filters: 32
Kernel Size: 3×3
Activation Function:
ReLU

This layer extracts low-level features such as edges and textures.

Max Pooling Layer

Pool Size: 2×2

Pooling reduces the spatial dimensions of the feature maps and decreases computational complexity.

Convolutional Layer 2

Filters: 64
Kernel Size: 3×3
Activation Function:
ReLU

This layer extracts more complex patterns from the feature maps.

Max Pooling Layer

Pool Size: 2×2

Further reduces the spatial dimensions of the data.

Flatten Layer

The flatten layer converts the 2D feature maps into a one-dimensional vector that can be processed by the fully connected layers.

Fully Connected Layer

Dense Layer: 128 neurons
Activation Function: ReLU

This layer learns high-level feature combinations.

Dropout Layer

Dropout is used to randomly deactivate neurons during training, which helps reduce overfitting.

Output Layer

Neurons: 10
Activation Function: Softmax

The softmax activation function outputs probability values for each class.

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6. Transfer Learning Using a Pretrained CNN

Transfer learning is a technique in which a neural network trained on a large dataset is reused for a different but related task.

In this experiment, a pretrained CNN model (MobileNetV2) was used for transfer learning. MobileNetV2 is a lightweight deep learning model designed for efficient image classification.

The pretrained model was used as a feature extractor, and new classification layers were added to adapt the model to the Fashion-MNIST dataset.

The following steps were performed:

1. Load the pretrained MobileNetV2 model without the top classification layer.
2. Freeze the pretrained layers to preserve learned features.
3. Add custom dense layers for classification.
4. Train the new layers using the Fashion-MNIST dataset.

Transfer learning improves model performance and reduces training time because the pretrained network has already learned general image features from large datasets.

7. Model Training

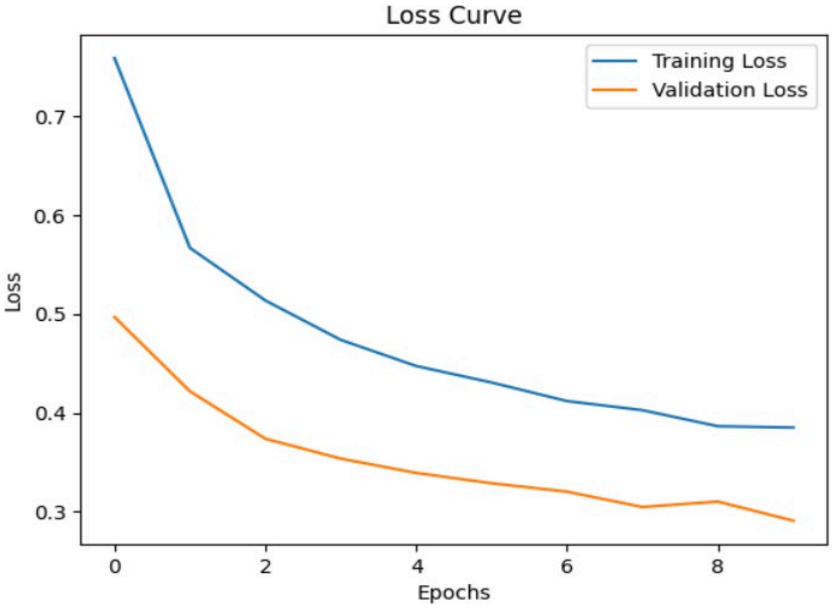
The CNN models were trained using the following parameters:

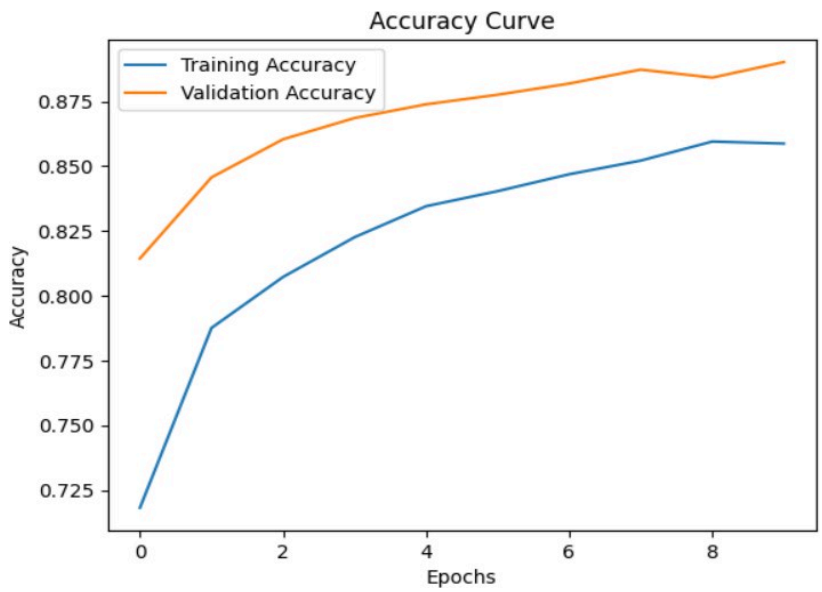
Optimizer: Adam
Loss Function: Categorical Cross-Entropy
Batch Size: 32
Epochs: 10

During training, the model continuously updates its weights to minimize the loss function. Both training accuracy and validation accuracy were monitored during training.

Two graphs were generated to analyze the learning process:

• Training vs Validation Loss
• Training vs Validation Accuracy





These graphs help determine whether the model is learning effectively and whether overfitting occurs.

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8. Model Evaluation

After training, the model was evaluated using the test dataset.

Several evaluation metrics were used:

Accuracy

Accuracy measures the percentage of correctly classified images.

Accuracy = Correct Predictions / Total Predictions

Precision

Precision measures how many predicted positive classifications were actually correct.

Recall

Recall measures the model's ability to correctly identify all relevant instances.

F1 Score

The F1 Score is the harmonic mean of precision and recall and provides a balanced evaluation of the model.

```
              precision    recall  f1-score   support

           0       0.86      0.83      0.84      1000
           1       0.99      0.98      0.99      1000
           2       0.85      0.82      0.83      1000
           3       0.88      0.90      0.89      1000
           4       0.81      0.83      0.82      1000
           5       0.97      0.99      0.98      1000
           6       0.67      0.69      0.68      1000
           7       0.94      0.96      0.95      1000
           8       0.97      0.98      0.98      1000
           9       0.97      0.94      0.96      1000

    accuracy                           0.89     10000
   macro avg       0.89      0.89      0.89     10000
weighted avg       0.89      0.89      0.89     10000
```

This experiment helped reinforce key concepts of deep learning including artificial neurons, activation functions, feedforward neural networks, optimization algorithms, and model evaluation techniques.

Future improvements could include hyperparameter tuning, cross-validation, and testing more advanced neural network architectures to further improve prediction performance.

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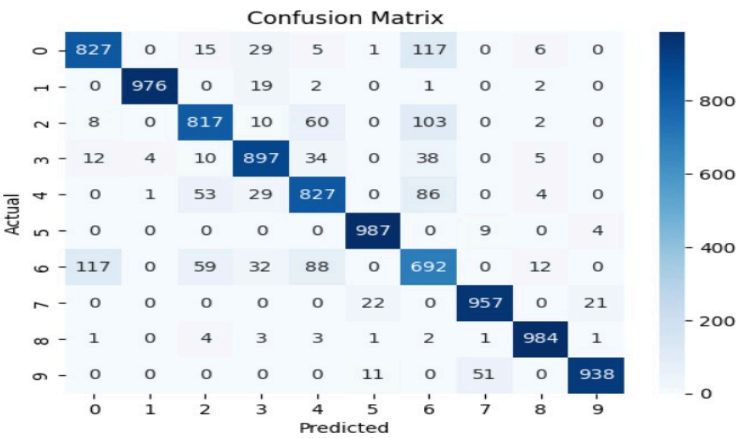
9. Confusion Matrix

A confusion matrix was used to visualize classification performance across all classes.

The confusion matrix shows:

True Positives
True Negatives
False Positives
False Negatives



This visualization helps identify which classes are being misclassified and provides insights into model weaknesses.

10. Comparison of Models

Two models were implemented and compared:

Model	Description
Custom CNN	Designed from scratch using convolution and pooling layers
Transfer Learning Model	Used pretrained MobileNetV2 for feature extraction

The transfer learning model typically achieved higher accuracy because it leveraged features learned from large-scale datasets.

11. Conclusion

In this experiment, Convolutional Neural Networks were implemented for multiclass image classification using the Fashion-MNIST dataset.

A custom CNN model was developed and trained from scratch. Data preprocessing and augmentation techniques were applied to improve model generalization. Additionally, transfer learning using the MobileNetV2 pretrained model was implemented.

The results demonstrated that transfer learning significantly improves model performance and training efficiency compared to training a CNN from scratch.

This assignment helped reinforce key concepts of deep learning, including convolution operations, pooling layers, activation functions, and transfer learning techniques used in modern computer vision systems.